Supporting Information:

Machine Learning for Hydrogen Storage Properties of TiFe Alloys Based on Literature-curated Dataset: Prediction and Limitation

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Table S1. The datasets used in this work and the corresponding sources

NO	Ti	Fe	Zr	V	Cr	Mn	Ni	Co	Y	T	Size1	Size2	Pa	Pd	wt	ref
1	1	0.85	0	0	0.15	0	0	0	0	25	45	50	0.06	0.04	1.38	[1]
2	1	0.8	0.05	0	0.15	0	0	0	0	25	45	50	0.032	0.02	1.42	[1]
3	1	0.75	0.1	0	0.15	0	0	0	0	25	45	50	0.027	0.02	1.17	[1]
4	1	0.7	0.15	0	0.15	0	0	0	0	25	45	50	0.018	0.018	0.9	[1]
5	1	0.9	0	0	0	0	0.1	0	0	50	270	830	0.2	0.15	1.7	[2]
6	1	0.85	0	0	0	0	0.15	0	0	50	270	830	0.09	0.08	1.86	[2]
7	1	0.8	0	0	0	0	0.2	0	0	50	270	830	0.03	0.02	1.86	[2]
8	1	0.9	0	0	0	0	0.1	0	0	20	270	830	0.045	0.035	1.7	[3]
9	1	0.9	0	0	0	0	0.1	0	0	30	270	830	0.1	0.06	1.7	[3]
10	1	0.9	0	0	0	0	0.1	0	0	40	270	830	0.15	0.13	1.7	[3]
11	1	0.9	0	0	0	0	0	0.1	0	20	270	830	0.25	0.13	1.8	[3]
12	1	0.9	0	0	0	0	0	0.1	0	30	270	830	0.35	0.21	1.76	[3]
13	1	0.9	0	0	0	0	0	0.1	0	40	270	830	0.5	0.3	1.7	[3]
14	1	0.9	0	0	0	0	0	0.1	0	50	270	830	0.82	0.43	1.7	[3]
15	1.1	0.8	0	0	0	0.2	0	0	0	40	48	72	0.325	0.183	1.8	[4]
16	1.1	0.8	0	0	0	0.2	0	0	0	60	48	72	0.514	0.327	1.86	[4]
17	1.1	0.8	0	0	0	0.2	0	0	0	80	48	72	0.721	0.511	1.77	[4]
18	1	0.8	0	0	0	0	0.2	0	0	40	48	75	0.0693	0.06	1.1	[5]
19	1	0.8	0	0	0	0	0.2	0	0	60	48	75	0.1254	0.1201	1.21	[5]
20	1	0.8	0	0	0	0	0.2	0	0	80	48	75	0.1949	0.194	1.22	[5]
21	1	0.77	0	0	0	0	0.2	0.03	0	40	48	75	0.0198	0.018	1.51	[5]
22	1	0.77	0	0	0	0	0.2	0.03	0	60	48	75	0.0377	0.0345	1.6	[5]
23	1	0.77	0	0	0	0	0.2	0.03	0	80	48	75	0.0658	0.062	1.6	[5]
24	1	0.75	0	0	0	0	0.2	0.05	0	40	48	75	0.0188	0.018	1.54	[5]
25	1	0.75	0	0	0	0	0.2	0.05	0	60	48	75	0.0356	0.0335	1.6	[5]
26	1	0.75	0	0	0	0	0.2	0.05	0	80	48	75	0.0634	0.06	1.62	[5]
27	1	0.7	0	0	0	0	0.2	0.1	0	40	48	75	0.0341	0.03	1.29	[5]
28	1	0.7	0	0	0	0	0.2	0.1	0	60	48	75	0.0605	0.06	1.31	[5]
29	1	0.7	0	0	0	0	0.2	0.1	0	80	48	75	0.1057	0.091	1.4	[5]
30	1.1	0.8	0	0	0	0.2	0	0	0	30	58	75	0.03	0.019	1.17	[6]
31	1.1	0.8	0	0	0	0.2	0	0	0	50	58	75	0.053	0.037	1.13	[6]
32	1.1	0.8	0	0	0	0.2	0	0	0	70	58	75	0.097	0.071	1.1	[6]
33	1.1	0.8	0	0	0	0.2	0	0	0	90	58	75	0.201	0.16	1.08	[6]
34	0.908	0.908	0.051	0	0	0.131	0	0	0	30	30	33	0.502	0.199	1.07	[7, 8]
35	0.908	0.908	0.051	0	0	0.131	0	0	0	40	30	33	0.7	0.329	1.05	[7, 8]
36	0.908	0.908	0.051	0	0	0.131	0	0	0	50	30	33	0.905	0.445	1.1	[7, 8]
37	0.908	0.908	0.051	0	0	0.131	0	0	0	60	30	33	1.164	0.576	1.06	[7, 8]
38	1	1	0	0	0	0	0	0.04	0	35	60	140	0.97	0.3	1.7	[9]
39	1	0.8	0	0	0	0.2	0	0	0	50	45	75	0.34	0.16	1.621	[10]
40	1	0.8	0	0	0	0.2	0	0	0	70	45	75	0.66	0.41	1.563	[10]
41	1	0.8	0	0	0	0.2	0	0	0	90	45	75	1.02	0.68	1.454	[10]
42	1.1	0.6	0.1	0	0	0.2	0.1	0	0	40	48	74	0.0459	0.0071	1.45	[11]
43	1.1	0.6	0.1	0	0	0.2	0.1	0	0	60	48	74	0.0746	0.0132	1.4	[11]
44	1.1	0.6	0.1	0	0	0.2	0.1	0	0	80	48	74	0.1146	0.0233	1.39	[11]
45	1.1	0.6	0.1	0	0	0.2	0.1	0	0	100	48	74	0.1717	0.0349	1.37	[11]
46	1.1	0.7	0.1	0	0	0.1	0.1	0	0	40	48	72	0.0091	0.0079	1.177	[12]
47	1.1	0.7	0.1	0	0	0.1	0.1	0	0	60	48	72	0.0161	0.0143	1.304	[12]
48	1.1	0.7	0.1	0	0	0.1	0.1	0	0	80	48	72	0.0259	0.0233	1.328	[12]
49	1.1	0.7	0.1	0	0	0.1	0.1	0	0	100	48	72	0.0383	0.0361	1.292	[12]
50	1	0.9	0	0	0	0	0	0	0	25	150	150	0.5	0.35	1.7	[13]

51	1	0.85	0	0	0	0.05	0	0	0	25	150	150	0.35	0.15	1.86	[13]
52	1	0.8	0	0	0	0.1	0	0	0	25	150	150	0.2	0.12	1.9	[13]
53	1	0.85	0	0	0	0.05	0	0	0	34	100	420	0.7	0.3	1.3	[14]
54	1	0.85	0	0	0	0.05	0	0	0	40	100	420	0.8	0.35	1.38	[14]
56	1	0.85	0	0	0	0.05	0	0	0	70	100	420	1.1	1	1.2	[14]
57	1	1	0	0	0	0	0	0	0	40	30	100	1.1	1	0.85	[15]
58	1	1	0	0	0	0	0	0	0	60	30	100	1.17	1.06	0.87	[15]
59	1	1	0	0	0	0	0	0	0	80	30	100	1.21	1.1	0.58	[15]
60	1.1	1	0	0	0	0	0	0	0	40	30	100	1.11	1	0.63	[15]
61	1.1	1	0	0	0	0	0	0	0	60	30	100	1.14	1.04	0.48	[15]
62	1.1	1	0	0	0	0	0	0	0	80	30	100	1.17	1.14	0.42	[15]
63	1.1	0.8	0	0	0	0.2	0	0	0	40	30	100	0.45	0.2	1.8	[15]
64	1.1	0.8	0	0	0	0.2	0	0	0	60	30	100	0.57	0.44	1.86	[15]
65	1.1	0.8	0	0	0	0.2	0	0	0	80	30	100	0.76	0.55	1.75	[15]
66	1	0.88	0	0	0	0.02	0	0	0	25	25	150	0.205	0.099	1.81	[16]
67	1	0.88	0	0	0	0.02	0	0	0	40	25	150	0.374	0.2	1.81	[16]
68	1	0.88	0	0	0	0.02	0	0	0	55	25	150	0.614	0.337	1.83	[16]
69	1	0.9	0	0	0	0	0	0	0	25	100	100	0.6675	0.3155	1.56	[17]
70	1	0.9	0	0.05	0	0	0	0	0	25	100	100	0.194	0.1125	1.66	[17]
71	1	0.9	0	0.1	0	0	0	0	0	25	100	100	0.134	0.092	1.74	[17]
72	1	0.8	0	0	0	0.1	0	0	0	25	100	100	0.1495	0.1005	1.6	[17]
73	1	0.8	0	0.05	0	0.1	0	0	0	25	100	100	0.0715	0.049	1.58	[17]
74	1	0.8	0	0.1	0	0.1	0	0	0	25	100	100	0.049	0.038	1.52	[17]
75	1	0.9	0	0	0	0.1	0	0	0	8	20	80	0.39	0.12	1.56	[18]
76	1	0.9	0	0	0	0.1	0	0	0	22	20	80	0.58	0.23	1.5	[18]
77	1	0.9	0	0	0	0.1	0	0	0	45	20	80	1.15	0.56	1.33	[18]
78	1	0.9	0	0	0	0.1	0	0	0	8	5	20	0.21	0.07	1.42	[18]
79	1	0.9	0	0	0	0.1	0	0	0	22	5	20	0.31	0.14	1.4	[18]
80	1	0.9	0	0	0	0.1	0	0	0	45	5	20	0.6	0.32	1.23	[18]
81	1	1	0	0	0	0	0	0	0	25	30	100	0.894	0.368	1.7	[19]
82	1	1	0	0	0	0	0	0	0	55	30	100	2.168	1.045	1.55	[19]
85	1	0.9	0	0	0	0	0	0	0	40	30	100	0.555	0.281	1.61	[19]
86	1	0.9	0	0	0	0	0	0	0	55	30	100	0.842	0.439	1.58	[19]
87	1	1	0	0	0	0.05	0	0	0	25	30	100	0.833	0.343	1.67	[19]
88	1	1	0	0	0	0.05	0	0	0	55	30	100	2.022	0.949	1.5	[19]
89	1	1	0	0	0	0.05	0	0	0	85	30	100	4.307	2.303	1.25	[19]
90	1	0.95	0	0	0	0.05	0	0	0	25	30	100	0.786	0.341	1.76	[19]
91	1	0.95	0	0	0	0.05	0	0	0	55	30	100	2.01	0.99	1.4	[19]
92	1	0.95	0	0	0	0.05	0	0	0	85	30	100	4.171	2.29	1.36	[19]
93	1	0.9	0	0	0	0.05	0	0	0	25	30	100	0.317	0.157	1.86	[19]
94	1	0.9	0	0	0	0.05	0	0	0	40	30	100	0.521	0.245	1.85	[19]
95	1	0.9	0	0	0	0.05	0	0	0	55	30	100	0.833	0.446	1.86	[19]
96	1	0.85	0	0	0	0.05	0	0	0	25	30	100	0.212	0.105	1.7	[19]
97	1	0.85	0	0	0	0.05	0	0	0	40	30	100	0.376	0.206	1.68	[19]
98	1	0.85	0	0	0	0.05	0	0	0	55	30	100	0.585	0.305	1.84	[19]
99	1	0.85	0	0	0	0.05	0	0	0	70	30	100	0.936	0.556	1.67	[19]
100	1	0.8	0	0	0	0.05	0	0	0	25	30	100	0.187	0.086	1.6	[19]
101	1	0.8	0	0	0	0.05	0	0	0	40	30	100	0.316	0.164	1.58	[19]
102	1	0.8	0	0	0	0.05	0	0	0	55	30	100	0.518	0.284	1.56	[19]
103	1	0.9	0	0	0	0.1	0	0	0	25	30	100	0.535	0.245	1.81	[19]
104	1	0.9	0	0	0	0.1	0	0	0	55	30	100	1.367	0.676	1.64	[19]
105	1	0.9	0	0	0	0.1	0	0	0	85	30	100	2.838	1.641	1.57	[19]
106	1	0.8	0	0	0	0.1	0	0	0	25	30	100	0.152	0.083	1.8	[19]
107	1	0.8	0	0	0	0.1	0	0	0	40	30	100	0.276	0.156	1.83	[19]

108	1	0.8	0	0	0	0.1	0	0	0	55	30	100	0.431	0.243	1.86	[19]
109	1	0.9	0	0	0.1	0	0	0	0	10	58	75	0.18	0.065	1.64	[20]
110	1	0.9	0	0	0.1	0	0	0	0	30	58	75	0.25	0.105	1.57	[20]
111	1	0.9	0	0	0.1	0	0	0	0	50	58	75	0.47	0.26	1.51	[20]
112	1	0.9	0	0	0	0.1	0	0	0	10	58	75	0.4	0.13	1.52	[20]
113	1	0.9	0	0	0	0.1	0	0	0	30	58	75	0.72	0.3	1.63	[20]
114	1	0.9	0	0	0	0.1	0	0	0	50	58	75	1.06	0.6	1.55	[20]
115	1.08	0.8	0	0	0	0.2	0	0	0.02	30	20	75	0.3	0.17	1.712	[21]
116	1.08	0.8	0	0	0	0.2	0	0	0.02	50	20	75	0.55	0.39	1.62	[21]
117	1.08	0.8	0	0	0	0.2	0	0	0.02	70	20	75	0.98	0.65	1.405	[21]
118	1.08	0.8	0	0	0	0.2	0	0	0.02	90	20	75	1.45	1.1	0.979	[21]
119	1.08	0.8	0.02	0	0	0.2	0	0	0.02	30	20	75	0.25	0.15	1.557	[21]
120	1.08	0.8	0.02	0	0	0.2	0	0	0.02	50	20	75	0.5	0.32	1.62	[21]
121	1.08	0.8	0.02	0	0	0.2	0	0	0.02	70	20	75	0.85	0.55	1.26	[21]
122	1.08	0.8	0.02	0	0	0.2	0	0	0.02	90	20	75	1.29	0.82	0.929	[21]
123	1.08	0.8	0.04	0	0	0.2	0	0	0.02	30	20	75	0.24	0.14	1.433	[21]
124	1.08	0.8	0.04	0	0	0.2	0	0	0.02	50	20	75	0.4	0.27	1.453	[21]
125	1.08	0.8	0.04	0	0	0.2	0	0	0.02	70	20	75	0.7	0.5	1.259	[21]
126	1.08	0.8	0.04	0	0	0.2	0	0	0.02	90	20	75	1.01	0.72	0.98	[21]
127	1.08	0.8	0.06	0	0	0.2	0	0	0.02	30	20	75	0.18	0.11	1.368	[21]
128	1.08	0.8	0.06	0	0	0.2	0	0	0.02	50	20	75	0.37	0.24	1.368	[21]
129	1.08	0.8	0.06	0	0	0.2	0	0	0.02	70	20	75	0.61	0.42	1.26	[21]
130	1.08	0.8	0.06	0	0	0.2	0	0	0.02	90	20	75	0.97	0.71	0.973	[21]
131	1.08	0.8	0.08	0	0	0.2	0	0	0.02	30	20	75	0.17	0.1	1.243	[21]
132	1.08	0.8	0.08	0	0	0.2	0	0	0.02	50	20	75	0.33	0.22	1.25	[21]
133	1.08	0.8	0.08	0	0	0.2	0	0	0.02	70	20	75	0.6	0.4	1.155	[21]
134	1.08	0.8	0.08	0	0	0.2	0	0	0.02	90	20	75	0.9	0.66	0.954	[21]
135	1.1	0.6	0.1	0	0	0.2	0.3	0	0	30	58	75	0.03	0.005	1.62	[22]
136	1.1	0.6	0.1	0	0	0.2	0.3	0	0	50	58	75	0.045	0.0097	1.6	[22]
137	1.1	0.6	0.1	0	0	0.2	0.3	0	0	70	58	75	0.06	0.013	1.56	[22]
138	1.1	0.6	0.1	0	0	0.2	0.3	0	0	90	58	75	0.08	0.02	1.54	[22]
139	1.08	0.6	0.1	0	0	0.2	0.3	0	0.02	30	58	75	0.02	0.0065	1.6	[22]
140	1.08	0.6	0.1	0	0	0.2	0.3	0	0.02	50	58	75	0.023	0.0095	1.58	[22]
141	1.08	0.6	0.1	0	0	0.2	0.3	0	0.02	70	58	75	0.03	0.014	1.57	[22]
142	1.08	0.6	0.1	0	0	0.2	0.3	0	0.02	90	58	75	0.046	0.02	1.43	[22]
143	1.06	0.6	0.1	0	0	0.2	0.3	0	0.04	30	58	75	0.0087	0.0056	1.78	[22]
144	1.06	0.6	0.1	0	0	0.2	0.3	0	0.04	50	58	75	0.02	0.008	1.7	[22]
145	1.06	0.6	0.1	0	0	0.2	0.3	0	0.04	70	58	75	0.031	0.023	1.63	[22]
146	1.06	0.6	0.1	0	0	0.2	0.3	0	0.04	90	58	75	0.045	0.03	1.6	[22]
147	1.04	0.6	0.1	0	0	0.2	0.3	0	0.06	30	58	75	0.02	0.003	1.6	[22]
148	1.04	0.6	0.1	0	0	0.2	0.3	0	0.06	50	58	75	0.023	0.008	1.57	[22]
149	1.04	0.6	0.1	0	0	0.2	0.3	0	0.06	70	58	75	0.025	0.015	1.55	[22]
150	1.04	0.6	0.1	0	0	0.2	0.3	0	0.06	90	58	75	0.04	0.021	1.47	[22]
151	1.02	0.6	0.1	0	0	0.2	0.3	0	0.08	30	58	75	0.017	0.004	1.65	[22]
152	1.02	0.6	0.1	0	0	0.2	0.3	0	0.08	50	58	75	0.024	0.01	1.62	[22]
153	1.02	0.6	0.1	0	0	0.2	0.3	0	0.08	70	58	75	0.03	0.017	1.6	[22]
154	1.02	0.6	0.1	0	0	0.2	0.3	0	0.08	90	58	75	0.061	0.045	1.57	[22]
155	1	0.86	0	0	0	0.1	0	0	0	45	38	48	0.5	0.41	1.91	[23]
156	0.95	1	0.05	0	0	0	0	0	0	25	25	150	0.62	0.23	1.25	[24]
157	1	0.95	0.05	0	0	0	0	0	0	25	25	150	0.31	0.14	1.41	[24]
158	1	1	0.05	0	0	0	0	0	0	25	25	150	0.54	0.19	1.32	[24]
159	1	0.86	0	0	0	0.07	0	0.07	0	25	380	550	0.57	0.2	1.82	[25]
160	1	0.86	0	0	0	0.07	0	0.07	0	35	380	550	0.7	0.3	1.83	[25]
161	1	0.86	0	0	0	0.07	0	0.07	0	45	380	550	1	0.4	1.82	[25]

162	1	0.86	0	0	0	0.07	0	0.07	0	55	380	550	1.04	0.53	1.8	[25]

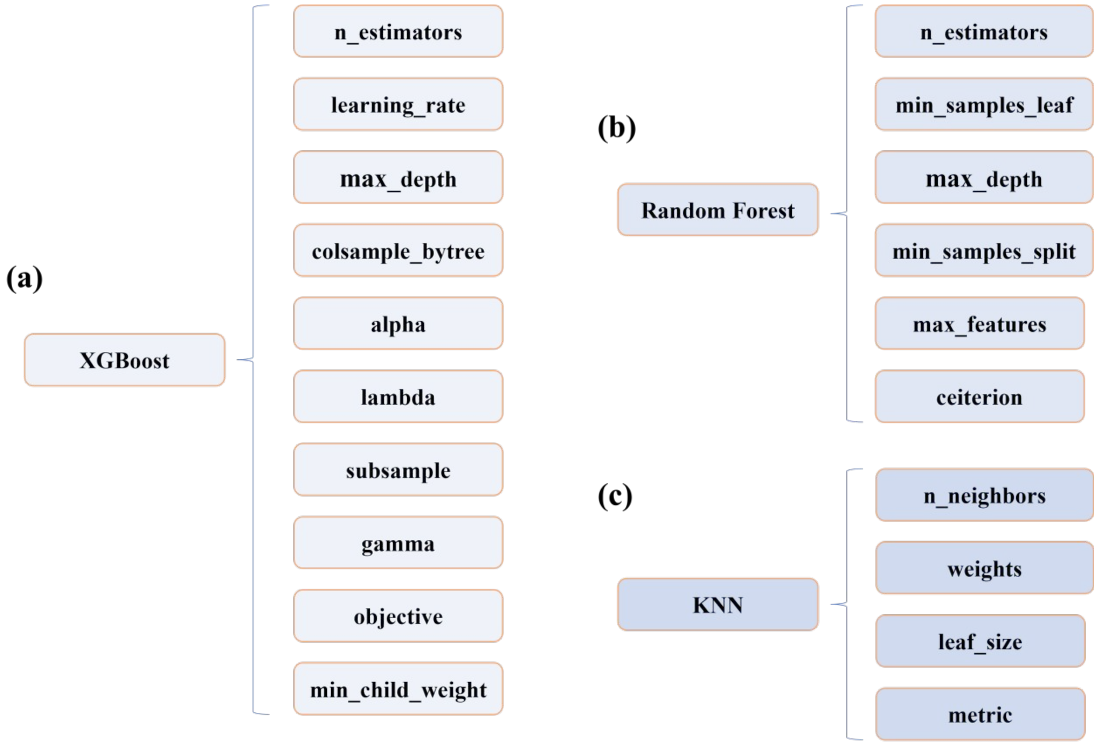

Figure S1. (a) Hyperparameters of the XGBoost model; (b) hyperparameters of the Random Forest model; (c) hyperparameters of the KNN model.

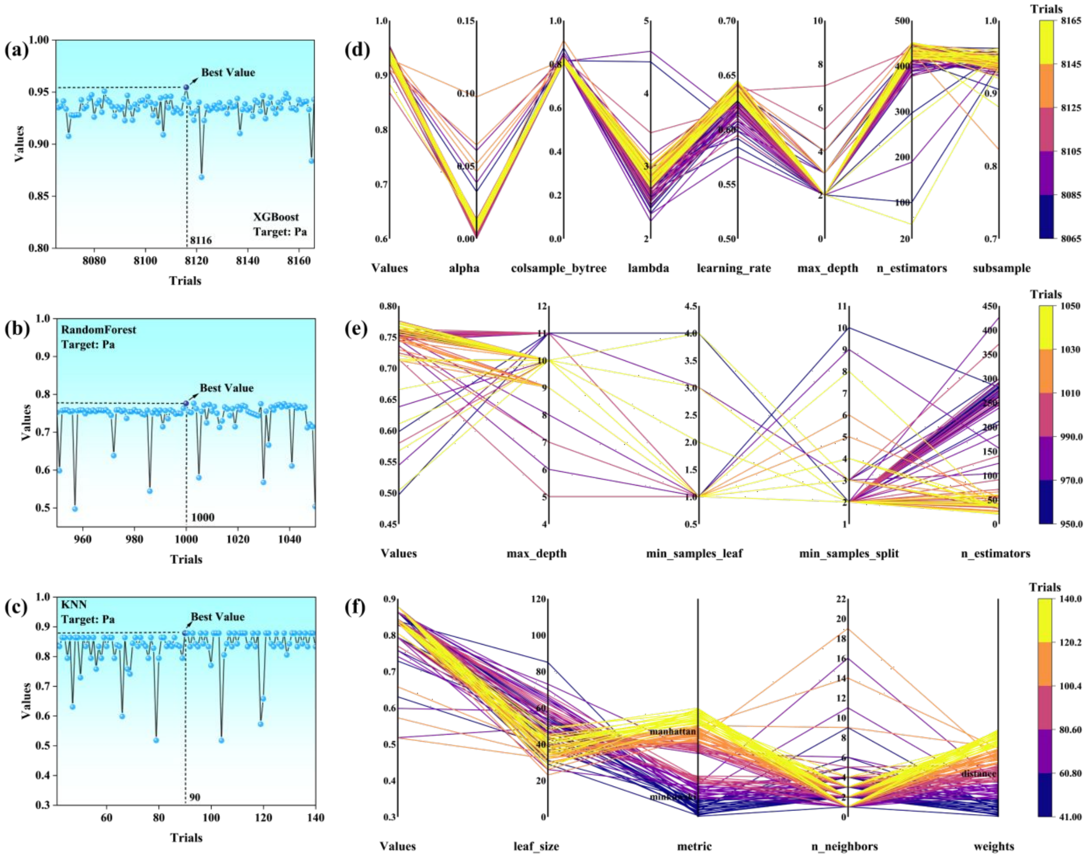

Figure S2. Plots of optimisation results of the three models for Pa; (a-c) Evaluation of optimisation results for Optuna; (d-f) Optimisation path diagrams for Optuna.

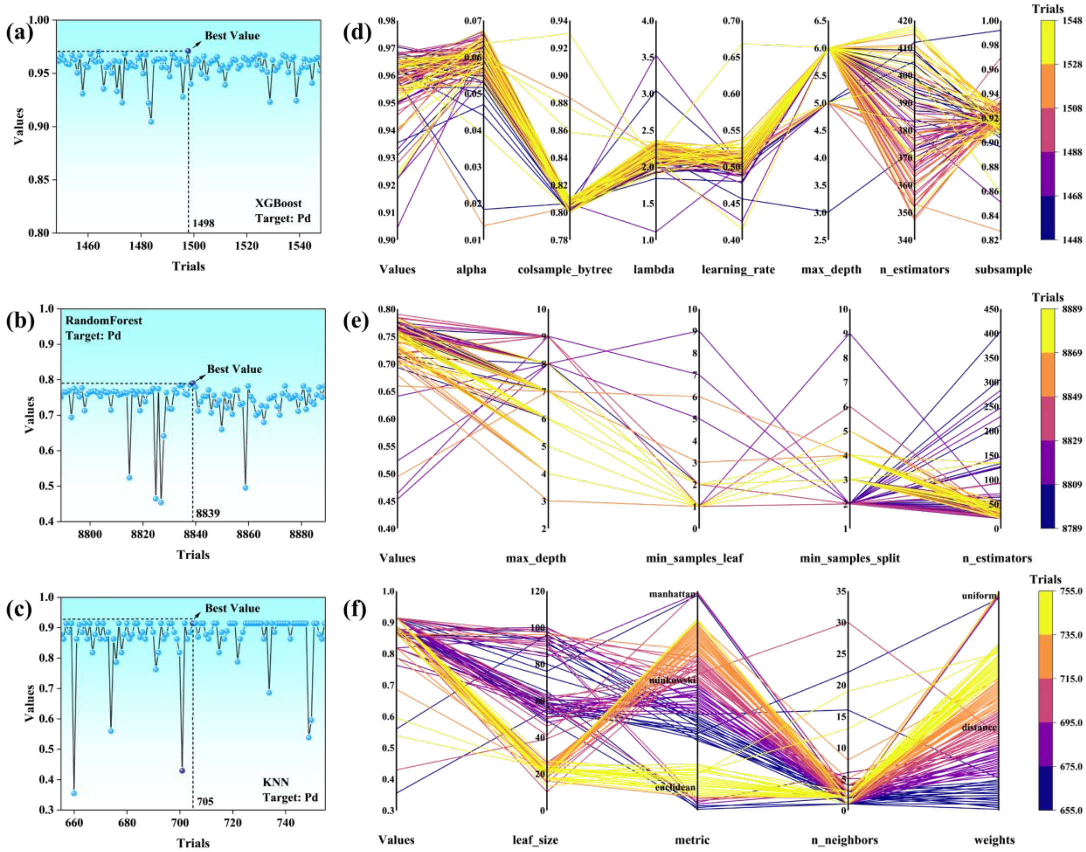

Figure S3. Plots of optimisation results of the three models for Pd; (a-c) Evaluation of optimisation results for Optuna; (d-f) Optimisation path diagrams for Optuna.

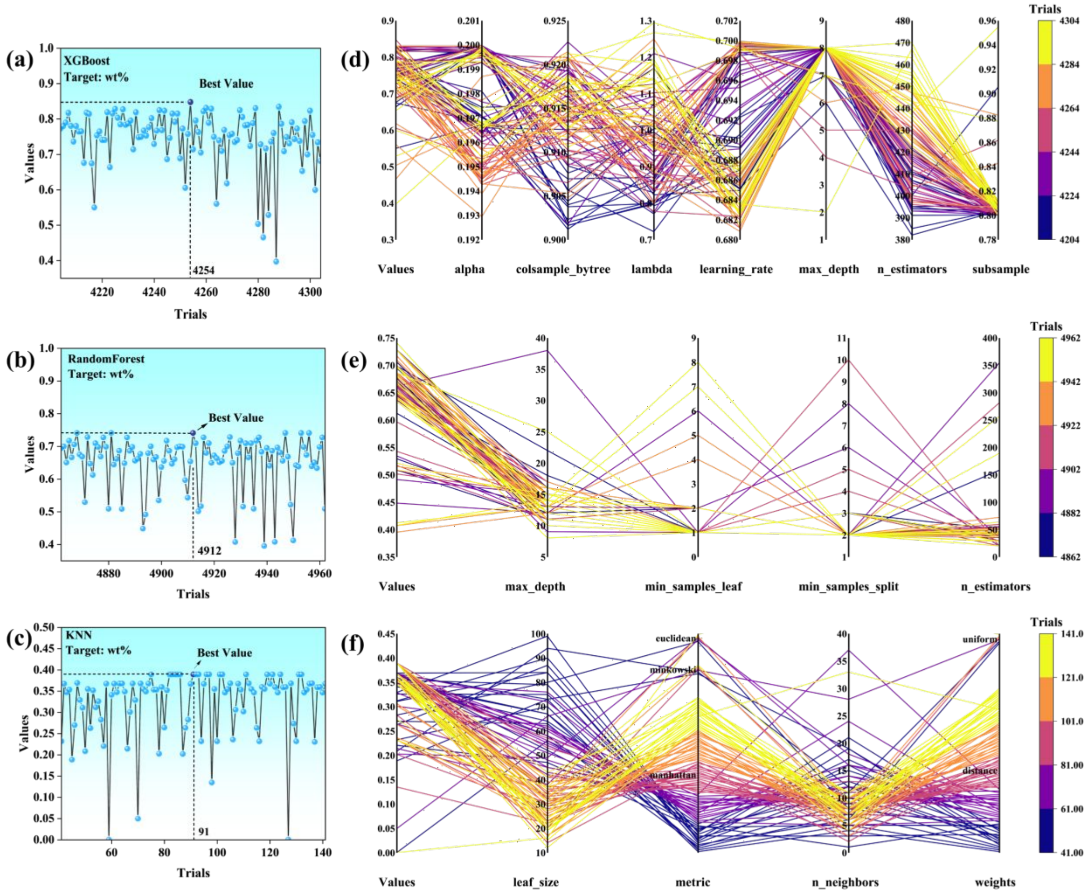

Figure S4. Plots of optimisation results of the three models for wt%; (a-c) Evaluation of optimisation results for Optuna; (d-f) Optimisation path diagrams for Optuna.

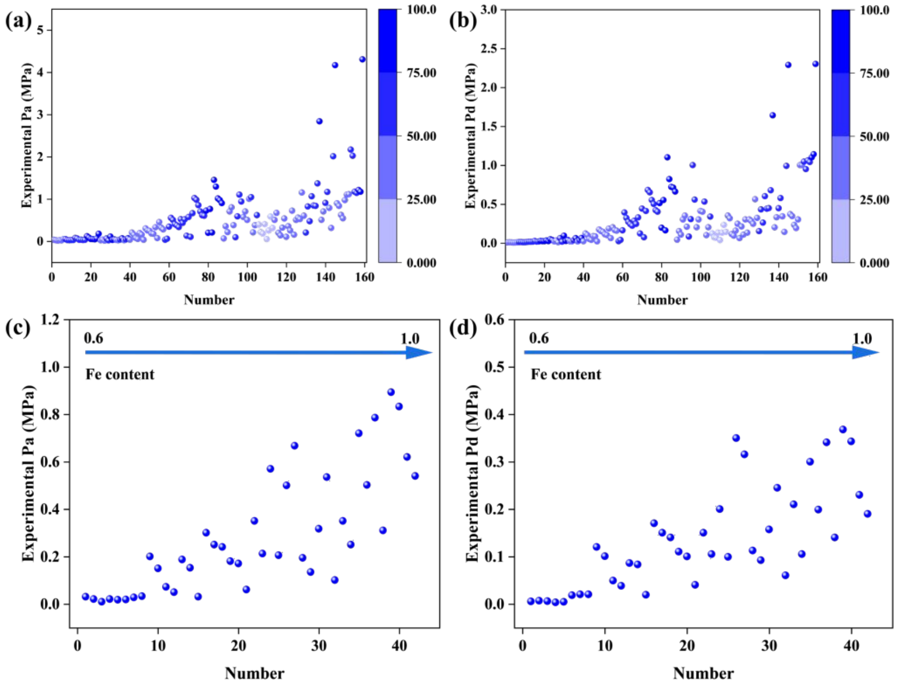

Figure S5. (a) Effect of change in Fe content on Pa; (b) Effect of change in Fe content on Pd; selected only at 25°C- 30°C, (c) Effect of change in Fe content on Pa; (d) Effect of change in Fe content on Pd.

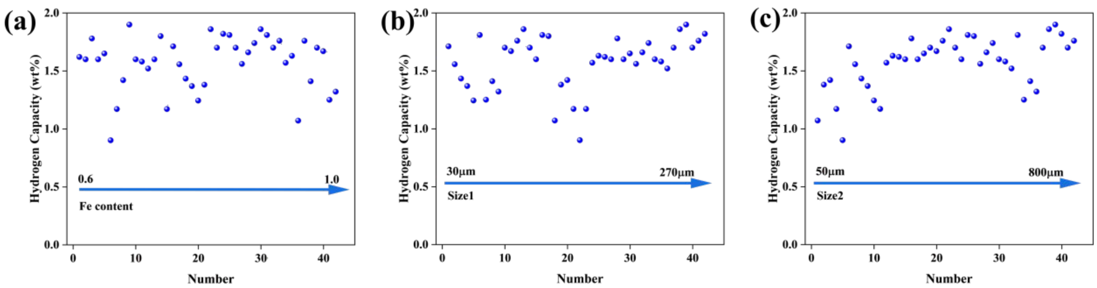

Figure S6. Selected experimental data at 25-30°C, (a) effect of Fe content; (b) effect of Size1; (c) effect of Size2.

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